

Circadian photoreception: The impact of light on human circadian rhythms

Jamie M. Zeitzer^{a,b,*} and Renske Lok^a

^a*Department of Psychiatry and Behavioral Sciences, Stanford University, Stanford, CA, United States*

^b*Mental Illness Research Education and Clinical Center, VA Palo Alto Health Care System, Palo Alto, CA, United States*

**Corresponding author: e-mail address: jzeitzer@stanford.edu*

Abstract

Light is the preeminent external influence in determining the position of the internal circadian clock relative to the outside world. In this chapter, we discuss the different parameters of light that impact how it influences the human circadian clock. We detail how the timing (phase), intensity, duration and temporal structure, and spectral composition all can modulate the impact of light on both the timing of the circadian clock as well as its amplitude. The neurobiological underpinnings of the system are briefly discussed in the context of understanding how light can evoke its observed effects on the circadian clock.

Keywords

Human, Circadian, Light, Dose response, Intensity, Phase, Mathematical modeling, Amplitude, Spectral sensitivity

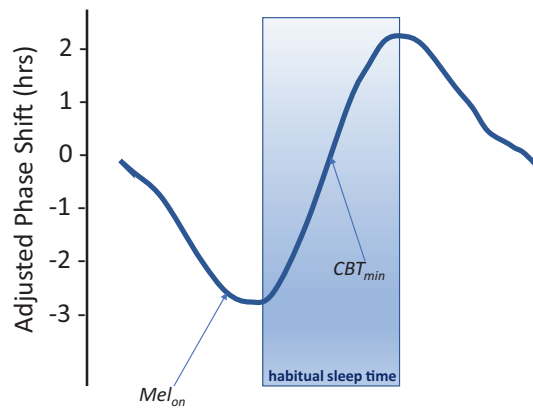
Light has a multitude of impacts on human brain function, notably allowing us to consciously perceive the visual environment. There are, however, a plethora of other aspects of brain function that are impacted by light, including synchronization of circadian rhythms, control of pupil size, altering mood and alertness, and changing neurohormonal output. The most commonly described of these are the impact of light on circadian timing, which will be the focus of this review. Many of these other aspects of light control of brain function are independently important (e.g., light influence on pupil size), but they are not necessarily driven by identical photoreceptive mechanisms. Many studies, however, have examined the impact of light non-circadian outputs (e.g., pupil, melatonin suppression, alertness) and attempted to impute how light act on the circadian system, which is a more difficult, lengthy, and expensive enterprise (Duffy and Dijk, 2002). It is critical to note that these other outputs are not

necessarily useful proxies if the main interest is in the response of the central circadian pacemaker (Zeitzer, 2018). In other words, the impact of light on the circadian timing system cannot necessarily be imputed from studies in which the circadian timing system is not actually being studied. This review will mainly focus on the impact of light on *circadian* function.

The circadian timing system is based around a central circadian clock that, in humans, is located in the suprachiasmatic nucleus of the hypothalamus (SCN) (Moore and Lenn, 1972; Stephan and Zucker, 1972). A circadian clock is an endogenous, near-24 h oscillator that is self-sustaining and can be synchronized to the outside world through zeitgebers (Gm., time cue) (Wever, 1979, 1992). Examples of behaviors that are influenced by circadian rhythms are the timing of sleep, pineal production of melatonin, and variation in cognitive function. In humans, as is the case in most mammals, the most potent zeitgeber is light. Light information gets from the outside world to the SCN via the intrinsically photosensitive retinal ganglion cells (ipRGC), the axons of which constitute the retinohypothalamic tract that innervates the SCN as well as many other hypothalamic nuclei (Gooley et al., 2003). The ipRGC themselves are sensitive to retinal light exposure through a combination of extrinsic input, from the rods and cones, and intrinsic input from melanopsin (Berson et al., 2002; Czeisler et al., 1995; Freedman et al., 1999). Other chapters in this volume will detail the neuroanatomy of this system in greater detail. Besides light, there are other zeitgebers that can influence the timing of the SCN, including melatonin (Lewy et al., 1992), social cues (Klerman et al., 1998), sleep (Deboer et al., 2004), food (Challet et al., 1998), and exercise (Baehr et al., 2003), though the direct magnitude of these effects are considerably smaller than that of light.

1 Phase response curve

The impact of light on circadian timing is dependent on the circadian time at which the light is received, a relationship that is codified as a “phase response curve” (PRC) (Fig. 1). Circadian time is the time of the internal circadian clock (SCN). In humans, this time (“phase”) is often imputed by the timing of the onset of melatonin secretion or the minimum of the circadian variation of core body temperature (Klerman et al., 2002). Determination of these times occur under specialized protocols and, thus far, cannot reliably be determined under ambulatory conditions due to masking of the circadian signal by common behaviors (Dijk and Duffy, 2020; Duffy and Dijk, 2002). The shape of a PRC is highly dependent on the specific stimulus (Johnson, 1990) but, in general, light exposure in the early part of the night causes the internal clock to slow down (delays: events will occur later on the next circadian cycle) while light exposure in the latter part of the night causes the circadian clock to speed up (advances: events will occur earlier on the next circadian cycle). Light exposure during the daytime has much smaller effects on circadian timing. Unlike many other mammalian species, there is no apparent “dead zone” in the human phase response curve, a time during which light exposure engenders no change in circadian timing (Khalsa et al., 2003). Light during the daytime, while having less direct impact on

**FIG. 1**

Phase-response relationship (PRC) between light intensity and circadian phase shift. Light in the late evening/early portion of the night elicits delays in timing of the circadian clock while light in the latter half of the night/early morning elicit advances. No area of non-responsiveness (dead zone) is evident in the human PRC.

Figure is adapted from data in Khalsa, S.B.S., Jewett, M.E., Cajochen, C., Czeisler, C.A., 2003. A phase-response curve to single bright light pulses in human subjects. J. Physiol. 549, 945–952.

circadian timing does engender potent changes to subsequent evening or overnight light exposure (Chang et al., 2011; Zeitzer et al., 2011). Greater amounts of daytime light exposure will mitigate the strength of nocturnal light exposure on changes in circadian timing. In essence, the magnitude of the phase shifting capacity of light appears to act as a relative, rather than absolute, stimulation of the circadian clock. While only limited data exist, this effect appears to not apply to morning light exposure if the morning light exposure is preceded by a night of sleep in presumed darkness (Zeitzer et al., 2011).

In addition to its phase shifting capacity, light exposure can also change the amplitude of the circadian clock. This is a mathematical construct for which there is no known physiologic correlate but may be related to the degree to which individual SCN neurons are synchronized with each other (An et al., 2013). Measurement of the amplitude of the circadian clock has been established through two markers—the fitted amplitude of core body temperature on a constant routine and the duration of melatonin secretion—and both as within person measurements (Czeisler et al., 1989; Shanahan et al., 1999). How light can impact both phase and amplitude simultaneously is most readily imagined as a clock, where the angular position is phase and the distance from the center (singularity) is amplitude (Winfrey, 2001). At lower amplitudes, phase changes are generally larger (Jewett et al., 1994). This two-dimensional representation can also explain the occurrence of Type 0 resetting in humans, in which very large (12h) phase shifts can occur with specific, multi-day sequences of light (Czeisler et al., 1989).

2 Duration sensitivity

The duration or continuity of light can also impact the magnitude of the light stimulation on circadian timing. Depending on the timing, longer duration exposures can lead to larger phase shifts, but the relative effectiveness (i.e., size of phase shift relative to length of exposure) greatly diminishes. Most laboratory studies of the impact of light on the human circadian clock use long, multi-hour exposure to engender large shifts (e.g., Figs. 1 and 2). However, pulses as short as 15 s can evoke significant changes in circadian timing (Rahman et al., 2017). While these shifts are smaller than that which are observed at longer durations (e.g., the 45-min delay observed after a 2-min pulse is 25% of the 3-h delay observed after a 6.5-h pulse (Gronfier et al., 2004; Rahman et al., 2017), the shorter pulses evoke an outsized impact on circadian timing (e.g., 22.7 h of circadian phase shift per hour of light for the 2-min pulse vs. 0.47 h of circadian phase shift per hour of light for the 6.5-h pulse, a nearly 50-fold difference in efficiency). The impact of short pulses of light can be further optimized by interleaving darkness between the light pulses during which, presumably, the cone photoreceptors are sensitized. Light flashes as short as 10 μ s, when given as a sequence over an hour with at least 15 s between flashes, can elicit changes in circadian timing (Joyce et al., 2022). When flashes are given with approximately 10 s of darkness between each flash, very large changes in circadian phase can be

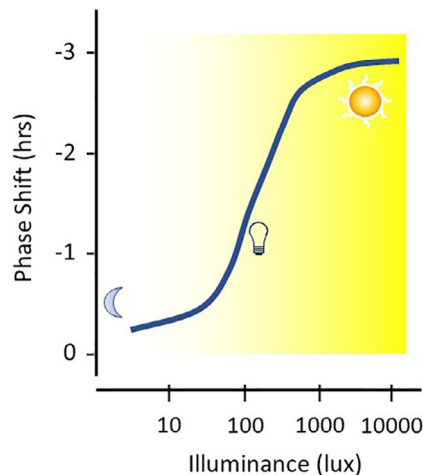


FIG. 2

Dose-response relationship between light intensity and circadian phase shift. Relatively little shift is seen with dim light under 10 lx, while saturated shifts are observed after exposure to very bright light. A log-linear relationship is observed in the range of normal room light.

Figure is adapted from data in Zeitzer, J.M., Dijk, D.-J., Kronauer, R.E., Brown, E.N., Czeisler, C.A., 2000. Sensitivity of the human circadian pacemaker to nocturnal light: melatonin phase resetting and suppression. *J. Physiol.* 526, 695–702.

evoked, approximate 2–3 h (cf., equiluminant light given for 60 min continuously evokes around 40 min of phase change) (Najjar and Zeitzer, 2016). In addition to these flash sequences evoking shifts 3–4 times larger than continuous light, they are >15,000 times more efficient than continuous light in terms of inducing a phase shift. A sequence of light flashes, however, is not a comfortable way to receive light while awake. Light flashes, however, can be delivered during sleep without interfering with sleep (Kaplan et al., 2019; Zeitzer et al., 2014), which is during the time at which the circadian system is usually most sensitive to light (PRC, Fig. 1).

The experiments describing the impact of varying temporal properties of light stimuli indicate that the circadian system appears to have a non-linear integration of light over time, as opposed to acting as a photon counter (Nelson and Takahashi, 1991). This integration process could occur at the level of the SCN (Meijer et al., 1993), but is more likely to occur at the level of the ipRGC (Berson et al., 2002). Recent data indicate that this integrative function of the ipRGC may be critical for the extrinsic cone input to the system (Wong and Fernandez, 2021) as melanopsin responses tend to be sluggish and active at higher illuminances (Berson et al., 2002; Wong et al., 2005, 2007). Understanding the manner in which light input is integrated over time is an important area in need of development. Most laboratory paradigms involve controlled, finite light exposures. Light exposure in the real world is, however, far more complicated (Webler et al., 2019) and current mathematical models do not accurately capture the impact of these patterns on circadian timing (Stone et al., 2019).

3 Intensity response curve

In addition to the temporal structure of light, the intensity of light is a major determinant of the magnitude of impact of light on circadian timing. For continuous light exposure, there is a sigmoidal (logistic) relationship between light intensity and its impact on circadian phase shifting (Fig. 2) (Zeitzer et al., 2000). At very low levels, below approximately 10 lx (very dim indoor lighting), there is relatively little impact of light on phase shifting. At very high levels, above approximately 1000 lx (dim outdoor light), the impact of light on circadian phase shifting is mostly saturated. Between these two illuminances, there is a log-linear relationship between increasing light intensity and the elicited circadian phase shift. This encompasses the normal range of indoor lighting, from dim bedrooms (<50 lx) to well-lit office spaces (500 lx). It is, however, important to note that this relationship was determined after multiple days of exposure to very dim light. As noted above, this type of pre-exposure to very dim light can enhance circadian responses to light and make the system more sensitive to light than it would be under real world circumstances (i.e., exposure to light intensities above 10 lx). It is like that under these circumstances, the dose-response curve to evening light is shifted to the right. Light exposure after sleep, during a mid-sleep awakening or in the morning, would likely express the sensitivity observed in the laboratory (Zeitzer et al., 2011). One source

of variance is interindividual variability in sensitivity to light, which has been observed for other non-image forming responses such as acute suppression of pineal melatonin production (Phillips et al., 2019), though the interindividual variability to the phase shifting effects of light has not been widely studied in healthy individuals. One other common source of variation is age. While healthy older individuals with unoccluded lenses have the same maximum circadian responses to light, they are less sensitive to light (the dose response curve is shifted to the right) (Duffy et al., 2007). Thus, to generate the same circadian phase shift in older adults, more light is needed. The amount of light needed to shift the average older adult, as opposed to the super healthy ones that were studied, is likely even greater as most have lens opacities that decrease light transmittance to the retina (Chylack et al., 1993). Systematic investigation of light sensitivity to phase shifting in children has not occurred. Indirect evidence from differences in light sensitivity of melatonin suppression across puberty (Crowley et al., 2015) would indicate that there may be an age-related difference in the phase shifting responses of adolescents and children to light, though this needs to be directly examined.

4 Spectral sensitivity

A final aspect of light to consider is the spectral content. The ipRGC have both intrinsic (melanopsin) and extrinsic (rods and cones) component inputs that vary in their spectral sensitivity. Melanopsin has a peak sensitivity around 480 nm (Bailes and Lucas, 2013), rods (rhodopsin) have a peak sensitivity around 505 nm, and cones (short-, medium-, and long-wavelength sensitive cone opsins) have a cumulative peak sensitivity around 550 nm (Wyszecki and Stiles, 1982). The reported peak sensitivity for the three-cone system, however, may be misleading in understanding cone input to the ipRGC as this spectral sensitivity function is derived from psychophysical experiments that presuppose a foveal dominance, which is unlikely the case for the cone input to the ipRGC given the nearly evenly spread distribution of the ipRGC in the retina (Esquivia et al., 2017; Gooley et al., 2001). The most thorough studies of non-image forming spectral sensitivity have been on the acute suppression of melatonin and indicate that this process is primarily if not entirely driven by signaling from intrinsic melanopsin, especially over longer durations of light exposure (Brainard et al., 2001; Brown, 2020; Gooley et al., 2010; Thapan et al., 2001). The ipRGC that are involved in light-mediated melatonin suppression, however, are quite possibly distinct from those involved in circadian phase shifting and may have a different combination of extrinsic inputs (Sonoda et al., 2018; Zeitzer, 2018). Extensive studies in rodents also indicate a likely role of cones in circadian phase shifting (Lucas et al., 2012). Unfortunately, there have been relatively few studies that have examined the spectral sensitivity of circadian phase shifting in humans (Gooley et al., 2010; Hanifin et al., 2019; Rahman et al., 2021; Smith and Eastman, 2009; Warman et al., 2003; Zeitzer et al., 1997). These limited studies would indicate that melanopsin alone cannot explain the human circadian phase

shifting response and that cones likely have a physiologically meaningful impact on circadian phase shifting, particularly with light stimuli that are of shorter durations and lower intensities (Gooley et al., 2010; Rahman et al., 2021; Wong and Fernandez, 2021). The impact of light spectrum on circadian phase shifting is, therefore, still a bit of a conundrum that will hopefully be solved in the coming years.

References

- An, S., Harang, R., Meeker, K., Granados-Fuentes, D., Tsai, C.A., Mazuski, C., Kim, J., Doyle III, F.J., Petzold, L.R., Herzog, E.D., 2013. A neuropeptide speeds circadian entrainment by reducing intercellular synchrony. *Proc. Natl. Acad. Sci. U. S. A.* 110, E4355.
- Baehr, E.K., Eastman, C.I., Revelle, W., Olson, S.H.L., Wolfe, L.F., Zee, P.C., 2003. Circadian phase-shifting effects of nocturnal exercise in older compared with young adults. *Am. J. Physiol.* 284, R1542–R1550.
- Bailes, H.J., Lucas, R.J., 2013. Human melanopsin forms a pigment maximally sensitive to blue light ($I_{\max} \approx 479$ nm) supporting activation of $G_{q/11}$ and $G_{i/o}$ signalling cascades. *Proc. Biol. Sci.* 280, 20122987.
- Berson, D.M., Dunn, F.A., Takao, M., 2002. Phototransduction by retinal ganglion cells that set the circadian clock. *Science* 295, 1070–1073.
- Brainard, G.C., Hanifin, J.P., Greeson, J.M., Byrne, B., Glickman, G., Gerner, E., Rollag, M.D., 2001. Action spectrum for melatonin regulation in humans: evidence for a novel circadian photoreceptor. *J. Neurosci.* 21, 6405–6412.
- Brown, T.M., 2020. Melanopic illuminance defines the magnitude of human circadian light responses under a wide range of conditions. *J. Pineal Res.* 69, e12655.
- Challet, E., Solberg, L.C., Turek, F.W., 1998. Entrainment in calorie-restricted mice: conflicting zeitgebers and free-running conditions. *Am. J. Physiol.* 274, R1751–R1761.
- Chang, A.M., Scheer, F.A., Czeisler, C.A., 2011. The human circadian system adapts to prior photic history. *J. Physiol.* 589, 1095–1102.
- Chylack, L.T., Wolfe, J.K., Singer, D.M., Leske, M.C., Bullimore, M.A., Bailey, I.L., Friend, J., McCarthy, D., Wu, S.-Y., 1993. The lens opacities classification system III. *Arch. Ophthalmol.* 111, 831–836.
- Crowley, S.J., Cain, S.W., Burns, A.C., Acebo, C., Carskadon, M.A., 2015. Increased sensitivity of the circadian system to light in early/mid-puberty. *J. Clin. Endocrinol. Metab.* 100, 4067–4073.
- Czeisler, C.A., Kronauer, R.E., Allan, J.S., Duffy, J.F., Jewett, M.E., Brown, E.N., Ronda, J.M., 1989. Bright light induction of strong (type 0) resetting of the human circadian pacemaker. *Science* 244, 1328–1333.
- Czeisler, C.A., Shanahan, T.L., Klerman, E.B., Martens, H., Brotman, D.J., Emens, J.S., Klein, T., Rizzo, J.F., 1995. Suppression of melatonin secretion in some blind patients by exposure to bright light. *N. Engl. J. Med.* 332, 6–11.
- Deboer, T., Van Steensel, M.J., D  t  ri, L., Meijer, J.H., 2004. Sleep states alter activity of suprachiasmatic nucleus neurons. *Nat. Neurosci.* 6, 1086–1090.
- Dijk, D.J., Duffy, J.F., 2020. Novel approaches for assessing circadian rhythmicity in humans: a review. *J. Biol. Rhythms* 35, 421–438.
- Duffy, J.F., Dijk, D.-J., 2002. Getting through to circadian oscillators: why use constant routines? *J. Biol. Rhythms* 17, 4–13.

- Duffy, J.F., Zeitzer, J.M., Czeisler, C.A., 2007. Decreased sensitivity to phase-delaying effects of moderate intensity light in older subjects. *Neurobiol. Aging* 28, 799–807.
- Esquivia, G., Lax, P., Pérez-Santonja, J.J., García-Fernández, J.M., Cuenca, N., 2017. Loss of melanopsin-expressing ganglion cell subtypes and dendritic degeneration in the aging human retina. *Front. Aging Neurosci.* 9, 79.
- Freedman, M.S., Lucas, R.J., Soni, B., Von Schantz, M., Muñoz, M., David-Gray, Z., Foster, R., 1999. Regulation of mammalian circadian behavior by non-rod, non-cone, ocular photoreceptors. *Science* 284, 502–504.
- Gooley, J.J., Lu, J., Chou, T.C., Scammell, T., Saper, C.B., 2001. Melanopsin in cells of origin of the retinohypothalamic tract. *Nat. Neurosci.* 4, 1165.
- Gooley, J.J., Lu, J., Fischer, D., Saper, C.B., 2003. A broad role for melanopsin in nonvisual photoreception. *J. Neurosci.* 23, 7093–7106.
- Gooley, J.J., Rajaratnam, S.M., Brainard, G.C., Kronauer, R.E., Czeisler, C.A., Lockley, S.W., 2010. Spectral responses of the human circadian system depend on the irradiance and duration of exposure to light. *Sci. Transl. Med.* 2, 31ra33.
- Gronfier, C., Wright Jr., K.P., Kronauer, R.E., Jewett, M.E., Czeisler, C.A., 2004. Efficacy of a single sequence of intermittent bright light pulses for delaying circadian phase in humans. *Am. J. Phys.* 287, E174–E181.
- Hanifin, J.P., Lockley, S.W., Cecil, K., West, K., Jablonski, M., Warfield, B., James, M., Ayers, M., Byrne, B., Gerner, E., Pineda, C., Rollag, M., Brainard, G.C., 2019. Randomized trial of polychromatic blue-enriched light for circadian phase shifting, melatonin suppression, and alerting responses. *Physiol. Behav.* 198, 57–66.
- Jewett, M.E., Kronauer, R.E., Czeisler, C.A., 1994. Phase-amplitude resetting of the human circadian pacemaker via bright light: a further analysis. *J. Biol. Rhythms* 9, 295–314.
- Johnson, C.H., 1990. *An Atlas of Phase Response Curves for Circadian and Circatidal Rhythms*. Department of Biology, Vanderbilt University.
- Joyce, D.S., Spitschan, M., Zeitzer, J.M., 2022. Duration invariance and intensity dependence of the human circadian system phase shifting response to brief light flashes. *Proc. Biol. Sci.* 9, 20211943.
- Kaplan, K.A., Mashash, M., Williams, R., Batchelder, H., Starr-Glass, L., Zeitzer, J.M., 2019. Effect of light flashes vs sham therapy during sleep with adjunct cognitive behavioral therapy on sleep quality among adolescents: a randomized clinical trial. *JAMA Netw. Open* 2, e1911944.
- Khalsa, S.B.S., Jewett, M.E., Cajochen, C., Czeisler, C.A., 2003. A phase-response curve to single bright light pulses in human subjects. *J. Physiol.* 549, 945–952.
- Klerman, E.B., Rimmer, D.W., Dijk, D.-J., Kronauer, R.E., Rizzo III, J.F., Czeisler, C.A., 1998. Nonphotic entrainment of the human circadian pacemaker. *Am. J. Physiol.* 274, R991–R996.
- Klerman, E.B., Gershengorn, H.B., Duffy, J.F., Kronauer, R.E., 2002. Comparisons of the variability of three markers of the human circadian pacemaker. *J. Biol. Rhythms* 17, 181–193.
- Lewy, A.J., Ahmed, S., Jackson, J.M.L., Sack, R.L., 1992. Melatonin shifts human circadian rhythms according to a phase-response curve. *Chronobiol. Int.* 9, 380–392.
- Lucas, R.J., Lall, G.S., Allen, A.E., Brown, T.M., 2012. How rod, cone, and melanopsin photoreceptors come together to enlighten the mammalian circadian clock. *Prog. Brain Res.* 199, 1–18.
- Meijer, J.H., Albus, H., Weidema, F., Ravesloot, J.H., 1993. The effects of glutamate on membrane potential and discharge rate of suprachiasmatic neurons. *Brain Res.* 603, 284–288.

- Moore, R.Y., Lenn, N.J., 1972. A retinohypothalamic projection in the rat. *J. Comp. Neurol.* 146, 1–9.
- Najjar, R.P., Zeitzer, J.M., 2016. Temporal integration of light flashes by the human circadian system. *J. Clin. Invest.* 126, 933–947.
- Nelson, D.E., Takahashi, J.S., 1991. Sensitivity and integration in a visual pathway for circadian entrainment in the hamster (*Mesocricetus auratus*). *J. Physiol.* 439, 115–145.
- Phillips, A.J.K., Vidafar, P., Burns, A.C., Mcglashan, E.M., Anderson, C., Rajaratnam, S.M.W., Lockley, S.W., Cain, S.W., 2019. High sensitivity and interindividual variability in the response of the human circadian system to evening light. *Proc. Natl. Acad. Sci. U. S. A.* 116, 12019–12024.
- Rahman, S.A., St Hilaire, M.A., Chang, A.M., Santhi, N., Duffy, J.F., Kronauer, R.E., Czeisler, C.A., Lockley, S.W., Klerman, E.B., 2017. Circadian phase resetting by a single short-duration light exposure. *JCI Insight* 2, e89494.
- Rahman, S.A., Brainard, G.C., Czeisler, C.A., Lockley, S.W., 2021. Spectral sensitivity of circadian phase resetting, melatonin suppression and acute alerting effects of intermittent light exposure. *Biochem. Pharmacol.* 191, 114504.
- Shanahan, T.L., Kronauer, R.E., Duffy, J.F., Williams, G.H., Czeisler, C.A., 1999. Melatonin rhythm observed throughout a three-cycle bright-light stimulus designed to reset the human circadian pacemaker. *J. Biol. Rhythms* 14, 237–253.
- Smith, M.R., Eastman, C.I., 2009. Phase delaying the human circadian clock with blue-enriched polychromatic light. *Chronobiol. Int.* 26, 709–725.
- Sonoda, T., Lee, S.K., Birnbaumer, L., Schmidt, T.M., 2018. Melanopsin phototransduction is repurposed by ipRGC subtypes to shape the function of distinct visual circuits. *Neuron* 99, 754–767.
- Stephan, F.K., Zucker, I., 1972. Circadian rhythms in drinking behavior and locomotor activity of rats are eliminated by hypothalamic lesions. *Proc. Natl. Acad. Sci. U. S. A.* 69, 1583–1586.
- Stone, J.E., Phillips, A.J.K., Ftouni, S., Magee, M., Howard, M., Lockley, S.W., Sletten, T.L., Anderson, C., Rajaratnam, S.M.W., Postnova, S., 2019. Generalizability of a neural network model for circadian phase prediction in real-world conditions. *Sci. Rep.* 9, 11001.
- Thapan, K., Arendt, J., Skene, D.J., 2001. An action spectrum for melatonin suppression: evidence for a novel non-rod, non-cone photoreceptor system in humans. *J. Physiol.* 535, 261–267.
- Warman, V.L., Dijk, D.-J., Warman, G.R., Arendt, J., Skene, D.J., 2003. Phase advancing human circadian rhythms with short wavelength light. *Neurosci. Lett.* 342, 37–40.
- Webler, F.S., Spitschan, M., Foster, R.G., Andersen, M., Peirson, S.N., 2019. What is the ‘spectral diet’ of humans? *Curr. Opin. Behav. Sci.* 30, 80–86.
- Wever, R.A., 1979. *The Circadian System of Man: Results of Experiments under Temporal Isolation*. Springer, New York.
- Wever, R.A., 1992. Basic principles of human circadian rhythms. In: Schmidt, T.F., Engel, B.T., Blümchen, G. (Eds.), *Temporal Variations of the Cardiovascular System*. Springer, Berlin, pp. 15–84.
- Winfree, A.T., 2001. *The Geometry of Biological Time*. Springer, New York.
- Wong, K.Y., Fernandez, F.X., 2021. Circadian responses to light-flash exposure: conceptualization and new data guiding future directions. *Front. Neurol.* 12, 627550.
- Wong, K.Y., Dunn, F.A., Berson, D.M., 2005. Photoreceptor adaptation in intrinsically photosensitive retinal ganglion cells. *Neuron* 48, 1001–1010.

- Wong, K.Y., Dunn, F.A., Graham, D.M., Berson, D.M., 2007. Synaptic influence on rat ganglion-cell photoreceptors. *J. Physiol.* 582, 279–286.
- Wyszecki, G., Stiles, W.S., 1982. *Color Science: Concepts and Methods, Quantitative Data and Formulae*. John Wiley and Sons, New York.
- Zeitzer, J.M., 2018. When is a proxy not a proxy? The foibles of studying non-image forming light. *J. Physiol.* 596, 2029–2030.
- Zeitzer, J.M., Kronauer, R.E., Czeisler, C.A., 1997. Photopic transduction implicated in human circadian entrainment. *Neurosci. Lett.* 232, 135–138.
- Zeitzer, J.M., Dijk, D.-J., Kronauer, R.E., Brown, E.N., Czeisler, C.A., 2000. Sensitivity of the human circadian pacemaker to nocturnal light: melatonin phase resetting and suppression. *J. Physiol.* 526, 695–702.
- Zeitzer, J.M., Friedman, L., Yesavage, J.A., 2011. Effectiveness of evening phototherapy for insomnia is reduced by bright daytime light exposure. *Sleep Med.* 12, 805–807.
- Zeitzer, J.M., Fiscaro, R.A., Ruby, N.F., Heller, H.C., 2014. Millisecond flashes of light phase delay the human circadian clock during sleep. *J. Biol. Rhythms* 29, 370–376.